

Given the following optimization problem, transform it in such a way as to obtain a minimization problem in which all decision variables must be non negative and all constraints are defined by lower inequalities.

$$\max -x_1^2 + \sin x_2$$

subject to

$$6x_1 - x_2^2 \geq 1,$$

$$x_1^2 + x_2^2 = 3,$$

$$x_1 \geq 2,$$

$$x_2 \in \mathbb{R}.$$